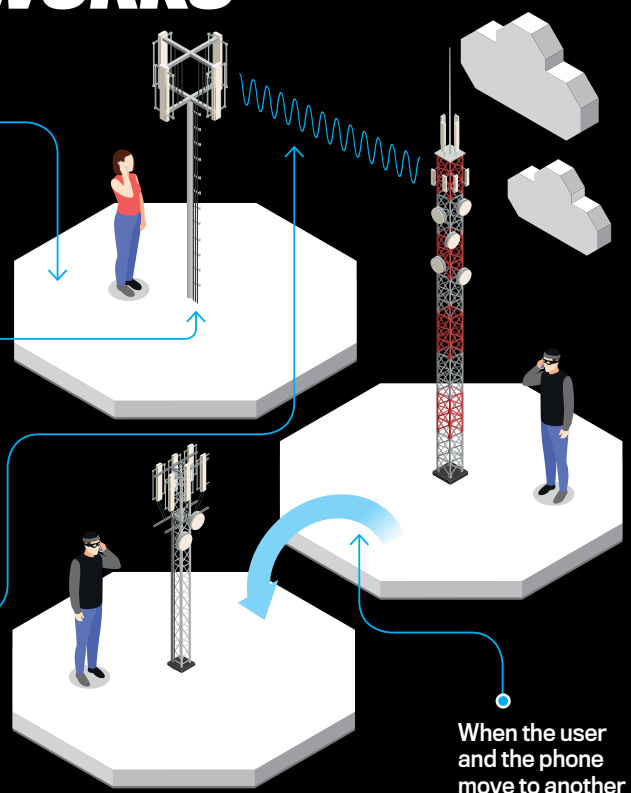
# HOW NETWORKS OPERATE

A mobile network is **divided into cells** where each phone is connected to exactly one cell at any given time (hence the name 'cell phone')

Each cell is operated by a cell tower on which a **wireless transceiver base station** is mounted. The base station maintains a wireless connection to all active phones in its cell

When a person speaks into her cell phone, **radio waves from it travel through the air** to a nearby cell tower, from where they are further transmitted



When the user and the phone move to another cell, the base stations perform an **exchange called a 'handover'**

Though **5G networks mostly rely on 4G and 4G LTE** (long-term evolution network, used to transmit wireless data at a faster rate compared to 3G) towers now, telcos are working to develop standalone 5G networks

These networks are based on a few core concepts such as:

- **Millimeter waves** or extremely high-frequency radio waves that give mobile devices gigabyte-plus speeds over short distances
- **Beamforming** or high-precision antenna arrays capable of directing wireless signals to individual devices
- **MIMO** (multiple-input, multiple-output) transmitters designed to deliver wireless data to devices at a much higher capacity

**While 4G uses radio towers to deliver both phone services and wireless internet to mobile devices, 5G incorporates new technology and higher radio frequencies. Both use more base stations to deliver faster speeds in a quicker response time**

While 4G uses radio towers to deliver phone services as well as wireless internet to mobile devices, 5G goes a step ahead and incorporates new technology and higher radio frequencies (see graphic, *How Networks Operate*).

## BENEFITS TO CONSUMERS

Shorn of the technical jargon, with 5G, downloads and streaming will happen without any delay. "The speed of data transmission will be at least 10 times faster than what we are experiencing in the 3G and 4G space," says K.G. Purushothaman, a partner with KPMG India. With the transition from 2G to 3G, the industry was already beginning to transform with better data speeds. "The transition from 3G to 4G was not very difficult. The data speed improved, the use cases (applications) of those improved speeds grew, and remote ways of working became the norm. With 5G, all this is going to increase exponentially," he adds.

According to Prashant Singhal, TMT (technology, media and telecom) emerging markets leader at EY, the big benefit for the end-customer will be that, effectively, there will be no latency or delay in the transfer of data. "This will mean immediate connectivity and you can interact in an immersive way as if you are talking to each other," he says. While 4G made it possible for HD videos to be played on mobile screens, 5G will support the era of 4K and 8K televisions that have ultra-high-definition viewing and many applications in the VR domain that require higher speeds, such as gaming.

How does 5G achieve all this? One of the major characteristics of 5G is its high bandwidth—the volume of information that can be sent over a connection in a specific amount of time. While 4G can manage only 200 megabytes of data per second (MBPS), 5G can handle up to 1 gigabyte or more. One megabyte (a data measurement unit) equals one million bytes whereas one gigabyte is one billion bytes. The other factor is low latency—the